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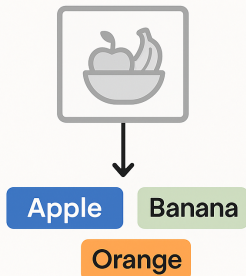
Generative AI

Ensemble Methodology and Evaluation

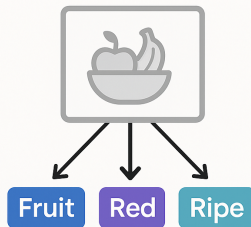
Introduction and Data Overview

Multi-class vs Multi-

Multi-Class Classification

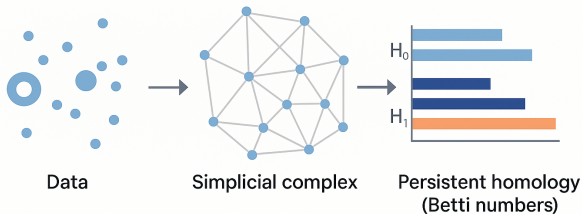


Multi-Label Classification



Homology

Homology in ML: From data to persistence



Modeling

FCN: Overview

Two types of models:

- **Original Data:** `sensor_data`
- **Extended Data:** all features available

Four types for each model:

- No Dropout / No Class Weight
- Dropout / No Class Weight
- No Dropout / Class Weight
- Dropout / Class Weight

FCN: Training & Hyperparameters

Random split 80/20

Grid-search on:

- Learning rate $\in \{10^{-3}, 5 \cdot 10^{-4}\}$
- Batch size $\in \{32, 64\}$
- Dropout probability $\in \{0.2, 0.3, 0.5\}$

FCN: Confusion Matrices

Confusion Matrix (Best Model: lr=0.0005, batch_size=32)

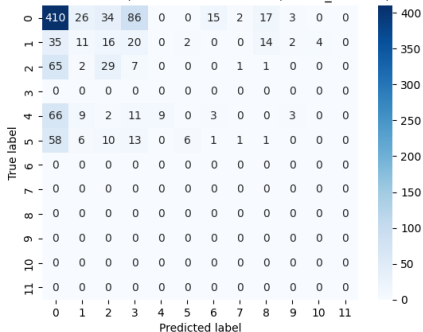


Figure: Original data (Dropout, no weight)

Confusion Matrix (FCD No Dropout, No Weights)

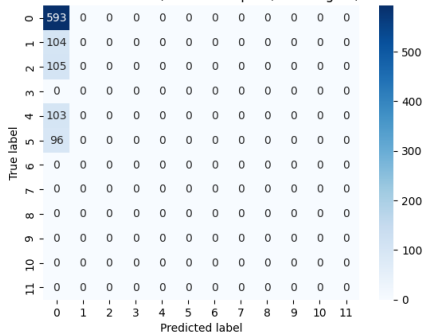
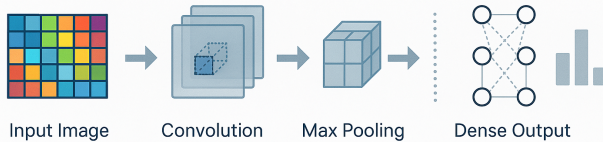


Figure: Extended data (all variants collapse)

FCN: Quantitative Comparison

Model	DO	CW	Acc. (%)	Macro-F1	Bal. Acc
Original	No	No	44.46	0.12	0.23
Original	Yes	No	46.45	0.13	0.24
Original	No	Yes	32.37	0.09	0.18
Original	Yes	Yes	29.57	0.10	0.17
Extended	No	No	59.24	0.15	0.20
Extended	No	Yes	59.24	0.15	0.20
Extended	Yes	No	59.24	0.15	0.20
Extended	Yes	Yes	59.24	0.15	0.20

2D Convolutional Neural Network: from pixels to features



Evaluated Model Families

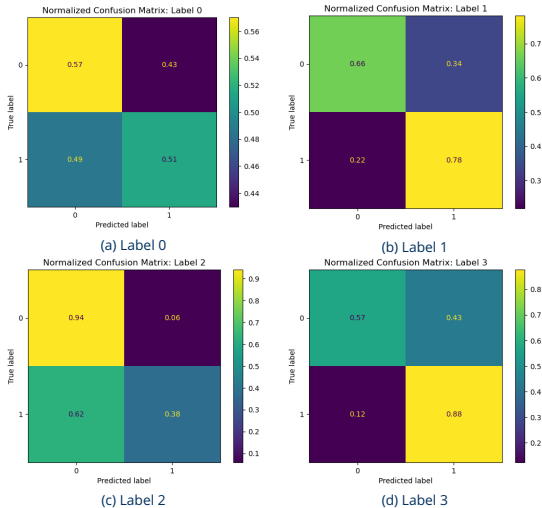
Model Family	Description
CNN	Simple 2-D Convolutional Neural Network (un-weighted baseline).
w-CNN	Weighted 2-D CNN using a weighted sampler with DVBAE (Deep Variational Bayes AutoEncoder).
w-ResNet	Weighted ResNet architecture, trained using the DVBAE-weighted sampler.
CNN-MC	Simple 2-D CNN applied to a multiclass classification setting.

2D CNN Results

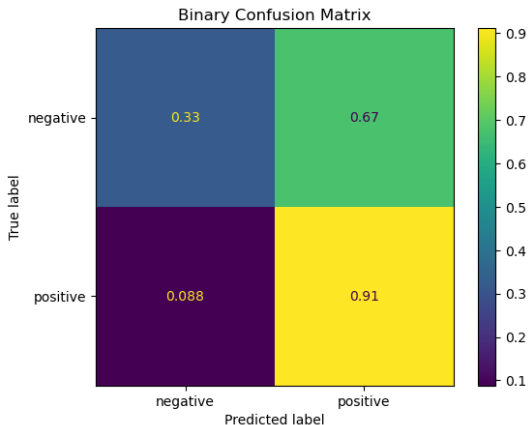
Architecture	Case	micro-F1	macro-F1	binary ACC	binary AUC
CNN	Standard 1	0.31	0.34	0.52	0.58
	Standard 2	0.28	0.32	0.47	0.59
	Extended 1	0.29	0.31	0.56	0.66
	Persistence 1	0.16	0.16	0.48	0.48
w-CNN	Standard 1	0.31	0.34	0.52	0.58
	Standard 2	0.28	0.29	0.49	0.65
	Extended 1	0.29	0.30	0.60	0.62
	Extended 2	0.28	0.31	0.47	0.59
	Extended 3	0.31	0.32	0.61	0.63
	Extended 4	0.28	0.31	0.58	0.64
	Persistence 1	0.12	0.13	0.49	0.48
w-ResNet	Standard 1	0.23	0.24	0.57	0.61

Table: Test scores of CNNs by architecture and case.

SILENT RAVEN

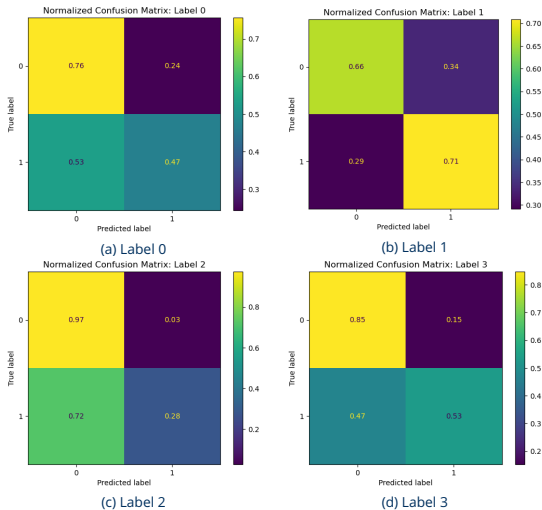


SILENT RAVEN

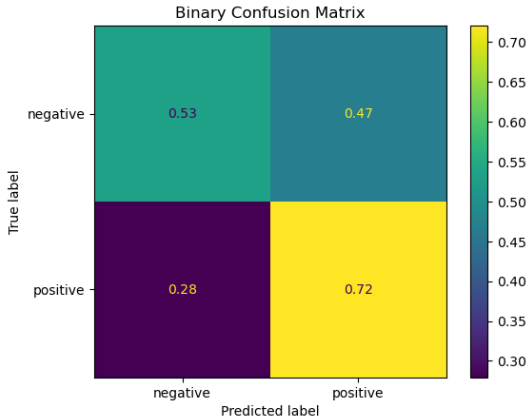


(a) Confusion Matrix binary case

VIVID NEBULA



VIVID NEBULA



(a) Confusion Matrix binary case

LSTM

Case No.	Model	Hyper-parameter tuning	Data Augmentation	Data Distribution	Multi label Classification	F1 score	Accuracy
1	LSTM	Random	No	As it is	No	0.3729	0.33
2	LSTM	Bayesian Optimization	No	As it is	No	0.079	0.3716
3	LSTM	Random	Yes	Label 0 = 1750 Other labels = 500	No	0.4500	0.4096
4	LSTM	Random	Yes	Label 0 = 1750 Other labels = 1250	No	0.4755	0.3766
5	LSTM	Random	Yes	Label 0 = 1750 Other labels = 1600	No	0.4565	0.3986
6	LSTM	Random	Under sampled	Label 0 = 500	No	0.3259	0.2867
7	LSTM	Random	No	As it is	Yes	front left : 0.03 , front right : 0.21 , rear left : 0.20 , rear right : 0.33	front left : 0.80 , front right : 0.87 , rear left : 0.79 , rear right : 0.83
8	LSTM	Random	No	As it is	Yes	front left : 0.02 , front right : 0.13 , rear left : 0.23 , rear right : 0.26	front left : 0.77 , front right : 0.82 , rear left : 0.81 , rear right : 0.64

Figure. Results of LSTM

Confusion Matrix: Case 4 (Highest F1 Score)

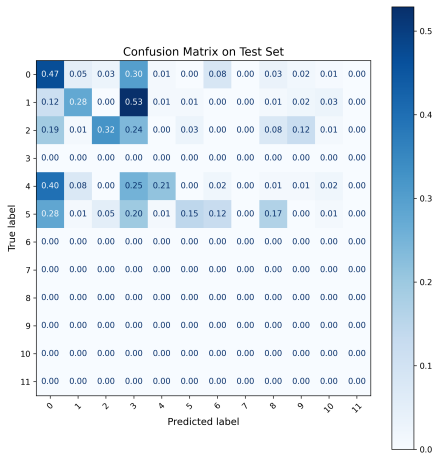


Figure. Normalized confusion matrix for Case 4.

Confusion Matrix: Case 7 (Highest Accuracy)

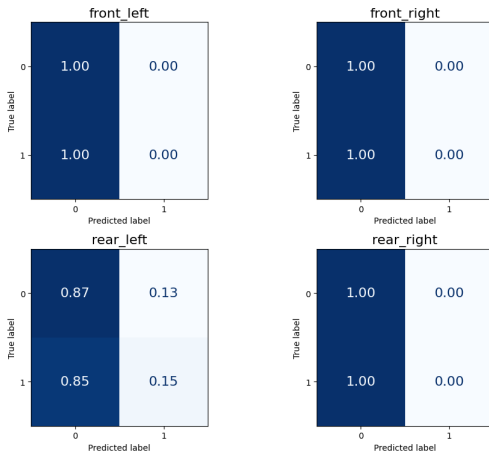


Figure. Normalized confusion matrix for Case 7.

Additional Models :Transformer Model

Case No.	Model	Hyper-parameter tuning	Data Augmentation	Data Distribution	Multi label Classification	F1 score	Accuracy
9	Transformer	Random	Yes	Label 0 = 1750 Other labels = 500	No	0.4240	0.3526
10	Transformer	Random	Yes	Label 0 = 1750 Other labels = 500	Yes	front left : 0.30 , front right : 0.11 , rear left : 0.13 , rear right : 0.15	front left : 0.81 , front right : 0.82 , rear left : 0.87 , rear right : 0.76

Figure: Performance of the Transformer model

Case 10 Results

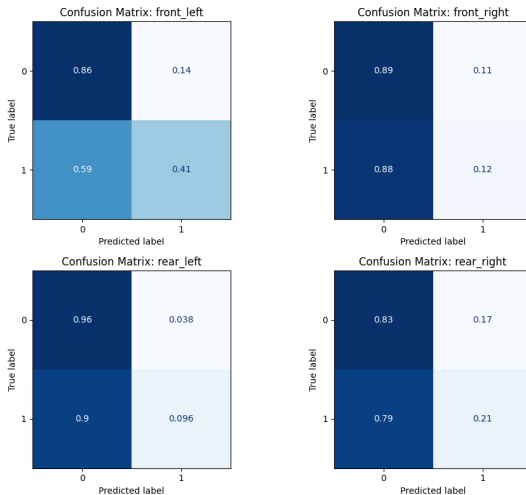


Figure: Results for Case 10.

Additional Models : XGBoost Results

Case No.	Model	F1 score	Accuracy
11	XGB with classifier chains	front left : 0.25 , front right : 0.33 , rear left : 0.36 , rear right : 0.23	front left : 0.62 , front right : 0.72 , rear left : 0.79 , rear right : 0.67

Figure: Prediction performance of the XGBoost model across all classes.

XGBoost Results: Confusion Matrix

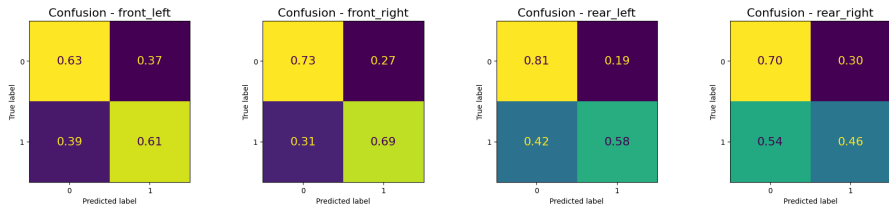


Figure: Normalized confusion matrix for the XGBoost model on the test set.

XGBoost Results: Binary Confusion Matrix

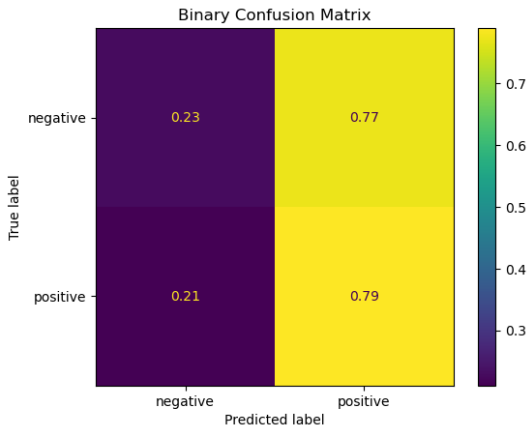


Figure: Normalized confusion matrix for the XGBoost model on the test set.

Model Comparison & Choice

- **FCN** – did not give promising results
- **CNN** (“Vivid Nebula”) – good binary balance, moderate per-label balance
- **XGBoost** – strong per-label balance, but biased to positive class in binary task

Generative AI

Goal

- Does synthetic data improve classifier performance?
- Try with:
 - Conditional GAN (CGAN)
 - Recurrent Neural Network (RNN)
- How to evaluate quality of models?

Challenge of Evaluation

- No objective loss function
 - e.g. Fréchet inception distance (FID) exists, but very complex to implement as it requires training a feature extractor model
- Therefore Manual inspection and downstream task performance used instead
- Created 2D animations of trajectories using Manim to visually assess output

Result

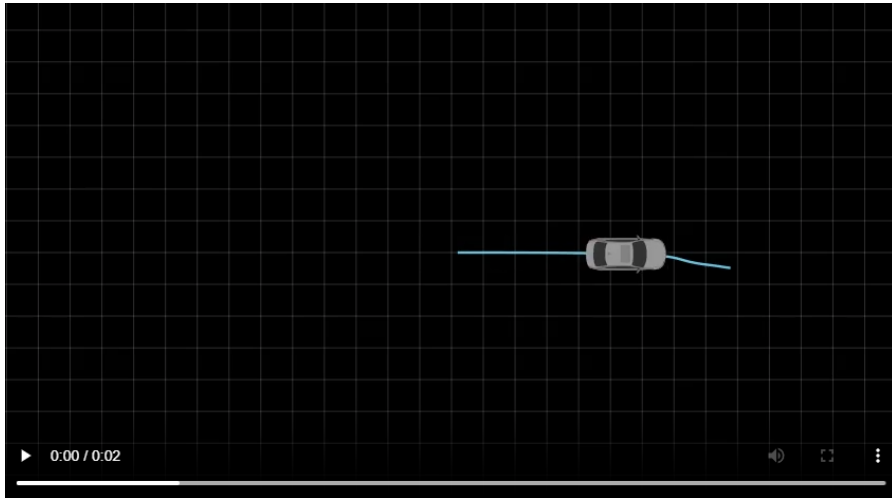


Figure: Screenshot of generated animation

CGAN Architecture

Discriminator:

$D_\phi(\text{label}, \text{braking data}) \rightarrow \mathbb{P}(\text{real and correct label})$

Trained on:

- (real label, real data)
- (fake label, real data)
- (real label, fake data)

Generator:

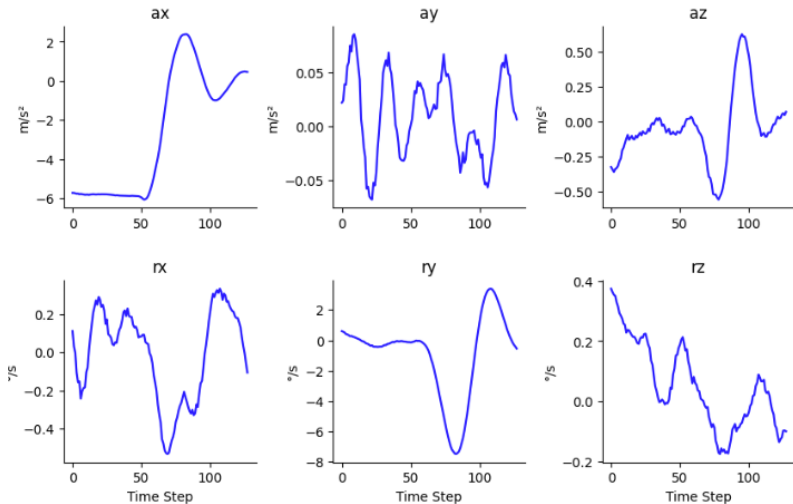
$G_\theta(\text{label}, \text{noise}) \rightarrow \text{braking data}$

Trained to fool the discriminator

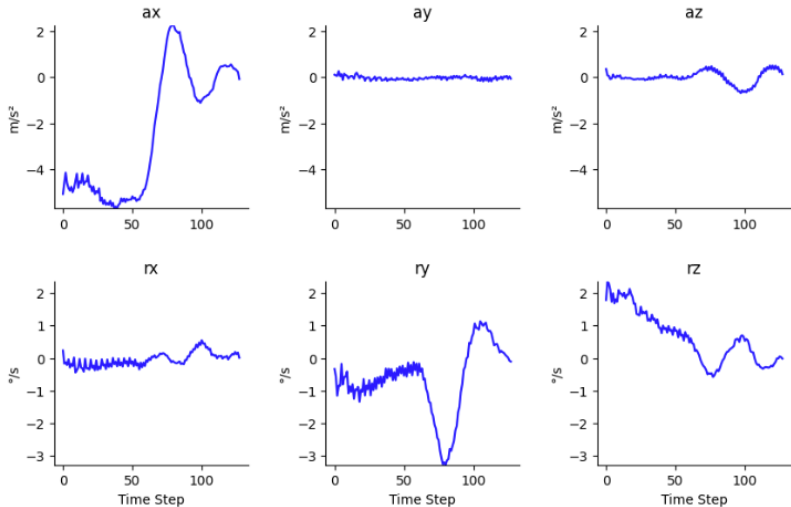
Training Techniques

- Instability due to Discriminator overpowering the Generator
- Solution: **Label Smoothing** for real data → reduce maximum certainty of the discriminator
- Architectures tested:
 - MLP for both Generator and Discriminator
 - CNN for Discriminator, DeconvNet for Generator

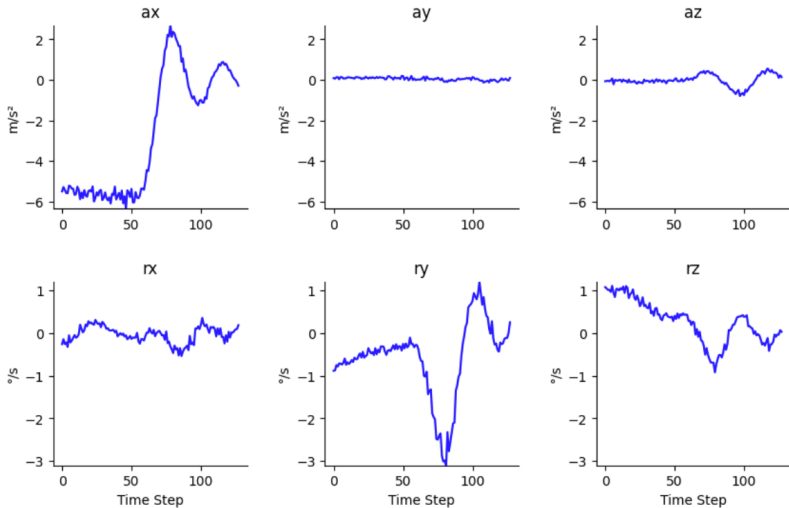
Real Data (Label 5)



CNN GAN (Label 5)



MLP GAN (Label 5)



RNN Architecture

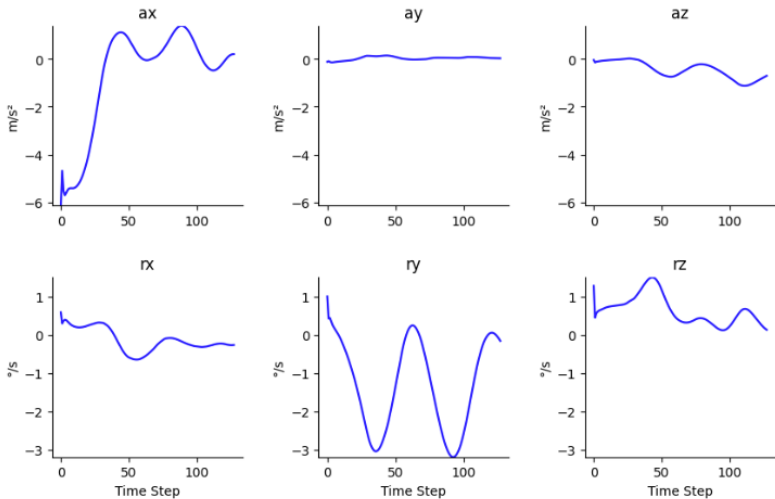
Model:

$$f_{\theta}(\text{label}, (x_0, x_1, \dots, x_{n-1})) \rightarrow (x_1, x_2, \dots, x_n)$$

Architecture Choice:

Uses LSTM instead of basic RNN because it worked better.

RNN Results (Label 5)

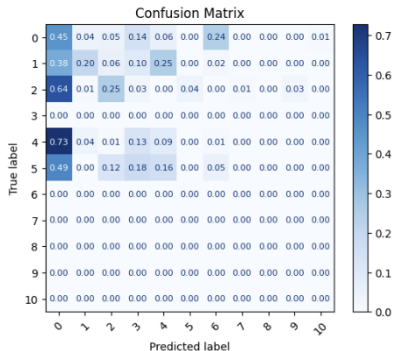


Data Extension

- Chose CNN GAN for generation.
- Synthetic data packaged with:
 - Median values of original car specs.
 - Label: 'Fake Car 9000'
- Labels generated uniformly → more balanced dataset.

LSTM with 100% real data

F1 score: 0.3610



LSTM with 50% real and 50% synthetic data

F1 score: 0.4306

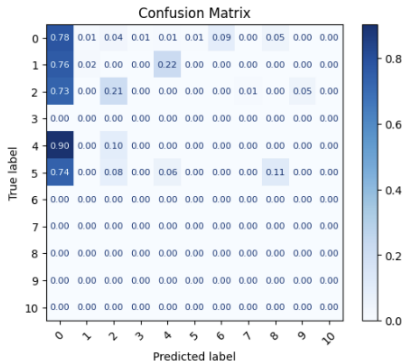


Figure: Performance improvement on some metrics is possible

LSTM with 100% synthetic data

F1 score: 0.0736

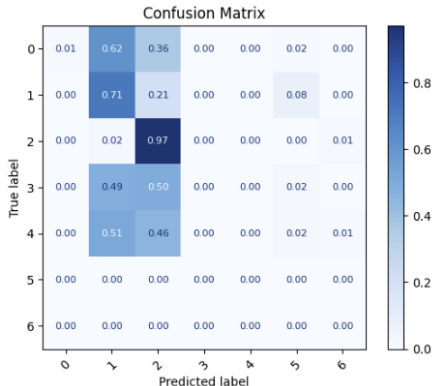


Figure: But fully synthetic data performs very badly

Conclusion

- Performance improvement on some metrics is possible
- Fully synthetic data performs very badly

Ensemble Methodology and Evaluation

Ensembles

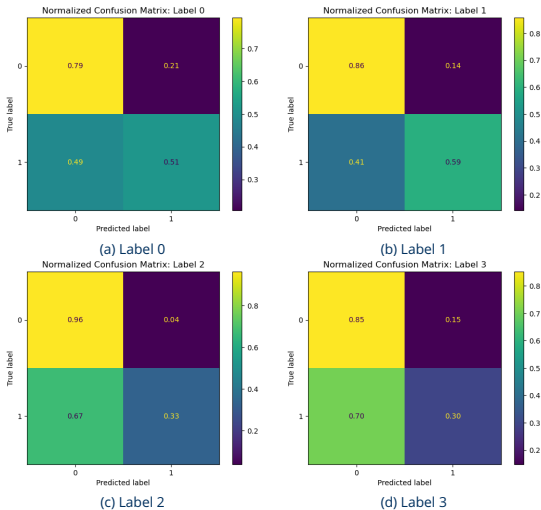
Table: Summary of ensemble candidates.

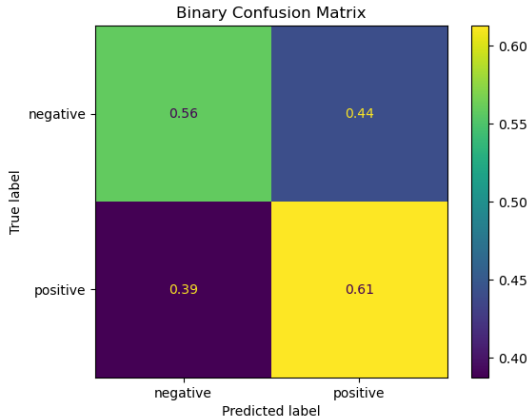
ID	Technique	Core Idea
E1	Calibrated Mean	Arithmetic mean of the two probability vectors after temperature scaling each base model.
E2	Label-Weighted Mean	Learn fixed weights $w_{\text{RF}}, w_{\text{CNN}}$ per label on the validation split and blend $p = w p_{\text{XGB}} + (1 - w) p_{\text{CNN}}$.
E3	Gated Selector	Linear regression outputs a gate $g \in [0, 1]$ per label and sample; final $p = g p_{\text{XGB}} + (1 - g) p_{\text{CNN}}$.
E4	Logit Stacking	Concatenate both probability vectors and train a logistic-regression meta-classifier.
E5	Dempster-Shafer Fusion	Interpret probabilities as mass functions, apply DS rules to combine evidence while discounting conflict.
E6	Rank Averaging	Convert probabilities to ranks per label, average ranks, then rescale to $[0, 1]$.

Ensembles Results

Table: Test-set performance of all ensemble methods.

Ensemble	Multilabel F1 (aggregate)		F1 per label				Binary		
	Micro	Samples	L0	L1	L2	L3	ACC	AUC	F1
E1	0.28	0.18	0.22	0.30	0.35	0.31	0.55	0.64	0.63
E2	0.29	0.18	0.20	0.32	0.40	0.32	0.46	0.55	0.63
E3	0.33	0.16	0.31	0.40	0.40	0.23	0.58	0.63	0.54
E4	0.29	0.19	0.21	0.31	0.40	0.32	0.46	0.63	0.63
E5	0.31	0.18	0.24	0.36	0.47	0.25	0.51	0.59	0.55
E6	0.26	0.20	0.22	0.29	0.27	0.24	0.56	0.66	0.59





(a) Confusion Matrix binary case